BS on answers → (min) max or (max) min

max Aggressive Cows → (min dist between cows) is max

C1 C2 ← C3

arr[] = [0 3 4 7 9 10] cows = 4

```
sort (arr, arr+n);
for ( i = 1 ; i <= max-min ; i++)
{
    if (canWePlace(arr, i, cows) == True)
        continue;
    else
        return (i-1);
}
```

TC →

BS-17. Aggressive Cows | Binary Search Hard

2.30

```
}

cntCows = 1, last = arr[0];

for( i=1 -> n-1)
{
    if (arr[i] - last >= dist)
    {
        cntCows ++;
        last = arr[i];
    }
}

if (cntCows >= cows) return true
else return false
}
```

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BS-17. Aggressive Cows | Binary Search Hard

2.30

return (i-1) ;

}

TC →

o(n^2) ← TC → O(max-min) × O(n)

SC → O(1)

bool canWePlace (arr, dist, cows)

{

cntCows = 1 , last = arr[0] ;

for (i = 1 → n-1)

{

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BS-17. Aggressive Cows | Binary Search Hard

2.30

arr[] = [0 3 4 7 9 10]   cows = 4

```
fn(arr, cows)
{
        sort(arr, arr+n);
    low=0, high = a[n-1] - a[0];
    while(low <= high)
    {
        mid = (low+high)/2
        if ( canWePlace(arr, mid, cows) == T)
                    ans = mid
                    low = mid+1;
        else
            high = mid-1;
    }
  ret high | ans ;
}
```

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BS-17. Aggressive Cows | Binary Search Hard

arr[] = [0 3 4 7 9 10]   cows = 4

```
fn(arr, cows)
{
        sort(arr, arr+n);
    low = 0, high = a[n-1] - a[0];

    while(low <= high)
    {
        mid = (low+high)/2

        if ( canWePlace(arr, mid, cows) == T)
                         ans = mid
                         low = mid+1;

        else
              high = mid-1;

    }
  ret high | ans ;
}
```

TC → N log N + { O(log₂( )) × O(N) }

SC → O(1)

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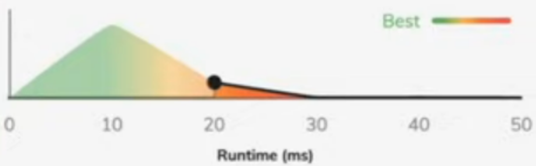
Correct Answer
few secs ago

Test Cases 10/10 | EXP: 80/80 | Penalty 0% | C++

Runtime graph | Memory graph

You performed better than **18.14%** | Runtime **20 ms**



Well done
You are in the highest level. Keep coding

Did you find these test cases useful?

Previous Submissions

All languages

Status | Test cases | EXP | Penalty | Languag

< Prev | Next >

C++ (g++ 5.4) | Autocomplete

```cpp
#include<bits/stdc++.h>
bool canWePlace(vector<int> &stalls, int dist, int cows) {
    int cntCows = 1, last = stalls[0];
    for(int i = 1;i<stalls.size();i++) {
        if(stalls[i] - last >= dist) {
            cntCows++;
            last = stalls[i];
        }
        if(cntCows >= cows) return true;
    }
    return false;
}
int aggressiveCows(vector<int> &stalls, int k)
{
    sort(stalls.begin(), stalls.end());
    int n = stalls.size();
    int low = 1, high = stalls[n-1] - stalls[0];
    while(low <= high) {
        int mid = (low + high) / 2;
        if(canWePlace(stalls, mid, k) == true) {
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }
    return high;
}
```

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